

These annotations were added after class, due to technical issues.\

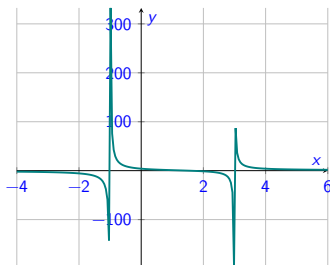
2627-MA140: Week 01, Lecture 3 (L03)

Polynomials and Rational Functions

Dr Niall Madden

University of Galway

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Today we rationalize the following topics:

- 1 News!
 - Tutorials
 - Tutorial sheet
- 2 Polynomials
 - Quadratic
 - Sketching polynomials
- 3 Rational Functions
 - Long division
- 4 Towards Partial Fractions
 - Quadratic factorization
- 5 Exercises

See also Sections 1.2 and 7.4(!)
of [https://math.libretexts.org/Bookshelves/Calculus/Calculus_\(OpenStax\)/01%3A_Functions_and_Graphs](https://math.libretexts.org/Bookshelves/Calculus/Calculus_(OpenStax)/01%3A_Functions_and_Graphs)

Slides are on canvas, and at
<https://www.niallmadden.ie/2627-MA140/>



Tutorials start **next** week. The schedule is:

Teams	Time	Venue	Leader
1–2	Tuesday 15:00	ENG- 2003	ST
3–4	Tuesday 15:00	ENG- 2034	DC
11–12	Thursday 11:00	ENG- 2002	ST
9–11	Thursday 11:00	ENG- 3035	DC
5–6	Friday 13:00	Eng- 2002	ST
7–8	Friday 13:00	Eng- 2035	DC

Do these data agree with information you have been provided?

Would you be interested to taking a tutorial through Irish? (Show of hands?)

You don't have to complete a graded assignment next week. However, this is a “practice” one available. See <https://universityofgalway.instructure.com/courses/58477/assignments/162051>

During tutorials, the tutor will solve some similar questions. You can access the **tutorial sheet** at the link above (and on Canvas: [Modules... Tutorial Sheets](#)).

The Tutorial Sheet has questions that are nearly identical to your own version.

Polynomials

Yesterday, we saw that...

Polynomials

Polynomials are functions of the form

$$y = f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x^1 + a_0, \quad x \in \mathbb{R},$$

where $a_0, a_1, \dots, a_n$ are real numbers called the **coefficients** of the polynomial. The number n is called the **degree** of the polynomial.

Examples: $y = x^3 - 2x^2 - 1$ is a cubic ($n=3$)

$y = \frac{1}{2} + 54.125 x^4 - x$ is a poly of degree $n=4$.

$y = e + \pi x$ is a poly of degree 1 (Linear).

But not $y = x^{-1} = \frac{1}{x}$ or $y = x^{1/2} = \sqrt{x}$

Example: quadratic

$x^2 - 2x - 3$ is a **quadratic** polynomial: it has degree $n = 2$.

There are many occasions when we want to **factorise** such quadratics, meaning we write them as the product of a pair of linear polynomials.

For example, we can **factorise** $x^2 - 2x - 3$ as

$$x^2 - 2x - 3 = (x - 3)(x + 1)$$

$$\begin{aligned} (x-3)(x+1) &= (x-3)x + (x-3)(1) \\ &= x^2 - 3x + x - 3 = x^2 - 2x - 3. \end{aligned}$$

Note: can see now that $x^2 - 2x - 3$
has zeros at $x = 3$, $x = -1$.

Example: quadratic

$x^2 - 2x - 3$ is a **quadratic** polynomial: it has degree $n = 2$.

There are many occasions when we want to **factorise** such quadratics, meaning we write them as the product of a pair of linear polynomials.

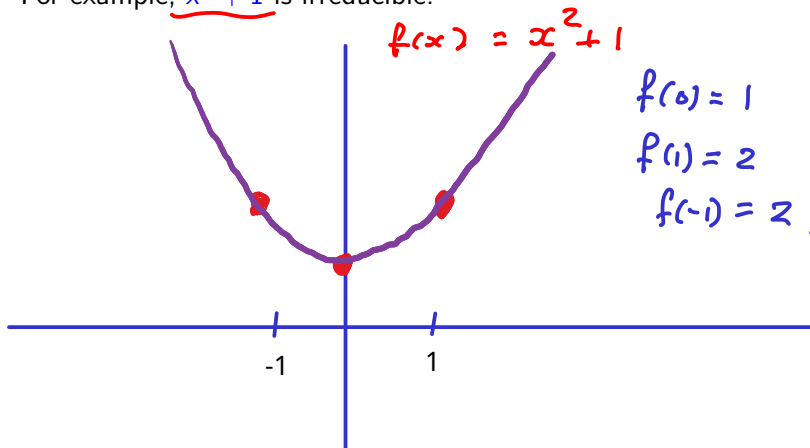
For example, we can **factorise** $x^2 - 2x - 3$ as

$$x^2 - 2x - 3 = (x - 3)(x + 1)$$

To sketch this: $f(x) = x^2 - 2x - 3$.
Use that $f(0) = -3$, $f(3) = 0$ $f(-1) = 0$.

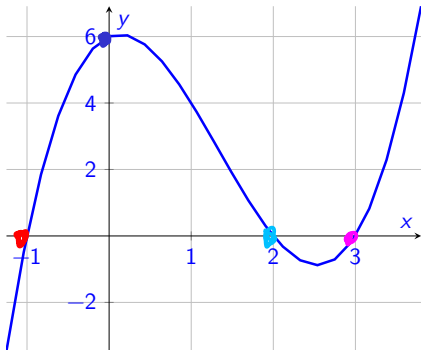
It is important to note that not all quadratic polynomials can be factorised as two linear polynomials. Such quadratics are called **irreducible**.

For example, $x^2 + 1$ is irreducible.



Example

$y = x^3 - 4x^2 + x + 6$ is a **cubic** function with degree $n = 3$.



Notes

$$f(x) = x^3 - 4x^2 + x + 6$$

$$f(0) = -6$$

$$f(-1) = (-1)^3 - 4(-1)^2 + (-1) + 6$$

$$= -1 - 4 - 1 + 6 = 0 \checkmark$$

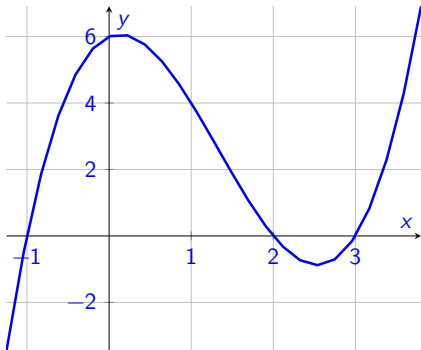
$$f(2) = 0 \quad f(3) = 0.$$

Check!

$$\text{So } f(x) = (x+1)(x-2)(x-3)$$

Example

$y = x^3 - 4x^2 + x + 6$ is a **cubic** function with degree $n = 3$.



Note: every cubic has at least one root, and thus at least one linear factor.

Fact

A polynomial function of degree n has **up to** $n-1$ turning points (“bends”).

Examples:

$n=0$: $f(x)=C$, a constant. This has no turning points.

$n=1$: $f(x)=ax+b$ (linear). That too has zero turning points.

$n=2$: $f(x) = ax^2 + bx + c$ (quadratic). This has 1 turning point.

$n=3$: $f(x) = x^3$

Although cubic, such as the one on slide 8 can have up to 2 turning points, this one has just one.

When sketching the graph of a function, we first find the **intercepts**:

- ▶ The **y-intercept** is where the graph of the function cuts the **y-axis**: found by letting $x = 0$.
 - ▶ The **x-intercepts** are where the function's graph cuts the **x-axis**. These points are also called the **roots** (or **zeros**). To find them, set **y** equal to zero and solve for **x**.
- * In addition, write down a table of some values of f . Plot the corresponding points, and the "join the dots".

Example

Sketch the graph of $y = -x^3 + x^2 + 2x$

1. Find y -intercept: $f(0) = 0$ ✓

2. To find the x -intercepts.

$$y = x(-x^2 + x + 2).$$

and find the zeros of $-x^2 + x + 2$.
check!

Factorizing we get $-x^2 + x + 2 = (-x + 2)(x + 1)$.

So $x = 2$ & $x = -1$ are also

x -intercepts (= "zeros" or "roots").

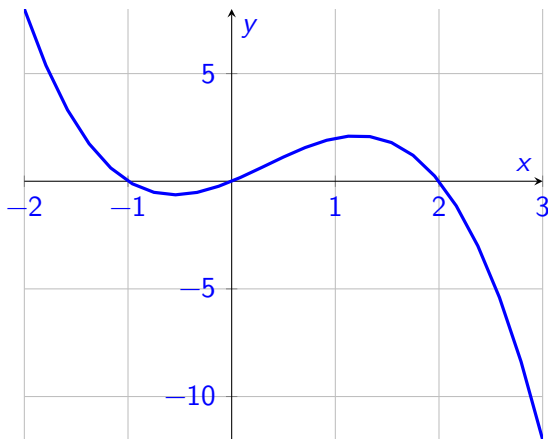
Also check
that

$$f(-2) = 8$$

$$f(1) = 2$$

$$f(3) = -12$$

Actual plot of $y = -x^3 + x^2 + 2x$



Rational Functions

Rational Functions

Rational Functions have the general form

$$f(x) = \frac{p(x)}{q(x)},$$

where $p(x)$ and $q(x)$ are polynomials.

► If degree of $p(x) <$ degree of $q(x)$,
 $f(x)$ is called a **strictly proper rational function**.

► If degree of $p(x) =$ degree of $q(x)$,
 $f(x)$ is called a **proper rational function**.

► If degree of $p(x) >$ degree of $q(x)$,
 $f(x)$ is called an **improper rational function**.

$$\begin{array}{r} x^2 + 1 \\ x^3 - 2x + \frac{1}{2} \end{array}$$

$$\begin{array}{r} x^2 \\ 2x^2 - 12 \end{array}$$

$$\begin{array}{r} x^2 + 1 \\ x - 1 \end{array}$$

Rational Functions

An improper or proper rational function can always be expressed as a polynomial plus a strictly proper rational function, for example by algebraic division.

Example

$$\frac{4x^3 + 4x^2 + 4}{x^2 - 3} = 4x + 4 + \frac{12x + 16}{x^2 - 3}$$

check: $4x + 4 = \frac{(4x + 4)(x^2 - 3)}{x^2 - 3} = \frac{4x^3 + 4x^2 - 12x - 12}{x^2 - 3}$.

Now use this to verify the equation above.

For the previous example, we can work this out ourselves using **Long Division** to divide numerator by denominator:

$$\begin{array}{r}
 4x + 4 \\
 x^2 - 3 \overline{) 4x^3 + 4x^2 + 4} \\
 \underline{- 4x^3 - 12x} \\
 4x^2 + 12x + 4 \\
 \underline{- 4x^2 - 12} \\
 12x + 16
 \end{array}$$

So $4x^3 + 4x^2 + 4 = (x^2 - 3)(4x + 4) + 12x + 16$

Which gives $\frac{4x^3 + 4x^2 + 4}{x^2 - 3} = 4x + 4 + \frac{12x + 16}{x^2 - 3}$

Example

Use long division to show that

$$\frac{3x^4 + 2x^3 - 5x^2 + 6x - 7}{x^2 - 2x + 3} = 3x^2 + 8x + 2 - \frac{14x + 13}{x^2 - 2x + 3}$$

Try this yourself!

Towards Partial Fractions

Rational Functions have the general form $f(x) = \frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials.

An (proper) rational function can often be written as a sum of simpler ones: **partial fractions**.

For example

$$\frac{8x - 12}{x^2 - 2x - 3}$$

can be written as

$$\frac{3}{x - 3} + \frac{5}{x + 1}$$

Check: (*next slide*)

Towards Partial Fractions

Towards Partial Fractions

In that last example, we manipulated $\frac{3}{x-3} + \frac{5}{x+1}$ to show that it is the **partial fraction** representation of $\frac{8x-12}{x^2-2x-3}$

In the next section, we'll work on the problem of *finding* the **partial fraction** representation of a rational function.

A key step is *factorization of polynomials*.

In the previous example, a key step was noting that $x^2 - 2x - 3 = (x - 3)(x + 1)$. That is, “ $x^2 - 2x - 3$ is factorized as $(x - 3)(x + 1)$ ”

Often, we'll want to factorize quadratics of the form

$$x^2 + bx + c,$$

meaning writing it as a product of two linear factors:

$$x^2 + bx + c = (x + q)(x + s).$$

To factorize, we need to find q and s such that

$$q + s = b \quad qs = c.$$

For example,

$$x^2 + 5x + 6 = (x + q)(x + r)$$

then $q + s = 5$ and $qs = 6$.

By inspection, we can see that we can take $q = 2$ and $s = 3$:

$$x^2 + 5x + 6 = (x + 2)(x + 3).$$

Factoring is useful because it makes it easy to find the roots:

$$(x + 2)(x + 3) = 0$$

implies

$$x = -2 \quad \text{or} \quad x = -3.$$

Exercise 1.3.1

Sketch the graphs of

(i) $y = 5x^2 - 7$

(ii) $y = x^2 - 4x + 3$

(iii) $y = x^3 - 6x^2 - 11x - 6$